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**Domain : AI/ML**

- **Dataset of Football team ranking :**

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
data=pd.read_csv("football.csv")
```

```
data
```

| | team | team_code | association | rank | previous_rank | points | previous_points | rated_matches |
|---|---|---|---|---|---|---|---|---|
| 0 | Argentina | ARG | CONMEBOL | 1 | 3 | 1876.118331 | 1874.814835 | 59 |
| 1 | Spain | ESP | UEFA | 2 | 2 | 1873.013187 | 1876.395199 | 56 |
| 2 | France | FRA | UEFA | 3 | 1 | 1869.428449 | 1877.322731 | 57 |
| 3 | England | ENG | UEFA | 4 | 4 | 1827.048678 | 1825.965482 | 57 |
| 4 | Portugal | POR | UEFA | 5 | 5 | 1766.177547 | 1763.834406 | 56 |
| ... | ... | ... | ... | ... | ... | ... | ... | ... |
| 206 | Bahamas | BAH | CONCACAF | 207 | 207 | 786.816217 | 786.816217 | 23 |
| 207 | US Virgin Islands | VIR | CONCACAF | 208 | 209 | 779.764266 | 779.764266 | 15 |
| 208 | British Virgin Islands | VGB | CONCACAF | 209 | 208 | 777.405734 | 782.137325 | 24 |
| 209 | Anguilla | AIA | CONCACAF | 210 | 210 | 760.251852 | 760.251852 | 18 |

- **Data Exploration :**

```
data.head()
```

| | team | team_code | association | rank | previous_rank | points | previous_points | rated_matches |
|---|---|---|---|---|---|---|---|---|
| 0 | Argentina | ARG | CONMEBOL | 1 | 3 | 1876.118331 | 1874.814835 | 59 |
| 1 | Spain | ESP | UEFA | 2 | 2 | 1873.013187 | 1876.395199 | 56 |
| 2 | France | FRA | UEFA | 3 | 1 | 1869.428449 | 1877.322731 | 57 |
| 3 | England | ENG | UEFA | 4 | 4 | 1827.048678 | 1825.965482 | 57 |
| 4 | Portugal | POR | UEFA | 5 | 5 | 1766.177547 | 1763.834406 | 56 |

Next steps: Generate code with data New interactive sheet

```
data.shape
```

```
(211, 8)
```

```
data.columns
```

```
Index(['team', 'team_code', 'association', 'rank', 'previous_rank', 'points',
       'previous_points', 'rated_matches'],
      dtype='object')
```

```python
data.dtypes
```

| | 0 |
|---|---|
| **team** | object |
| **team_code** | object |
| **association** | object |
| **rank** | int64 |
| **previous_rank** | int64 |
| **points** | float64 |
| **previous_points** | float64 |
| **rated_matches** | int64 |

**dtype:** object

```python
data.isnull().sum()
```

| | 0 |
|---|---|
| **team** | 0 |
| **team_code** | 0 |
| **association** | 0 |
| **rank** | 0 |
| **previous_rank** | 0 |
| **points** | 0 |
| **previous_points** | 0 |
| **rated_matches** | 0 |

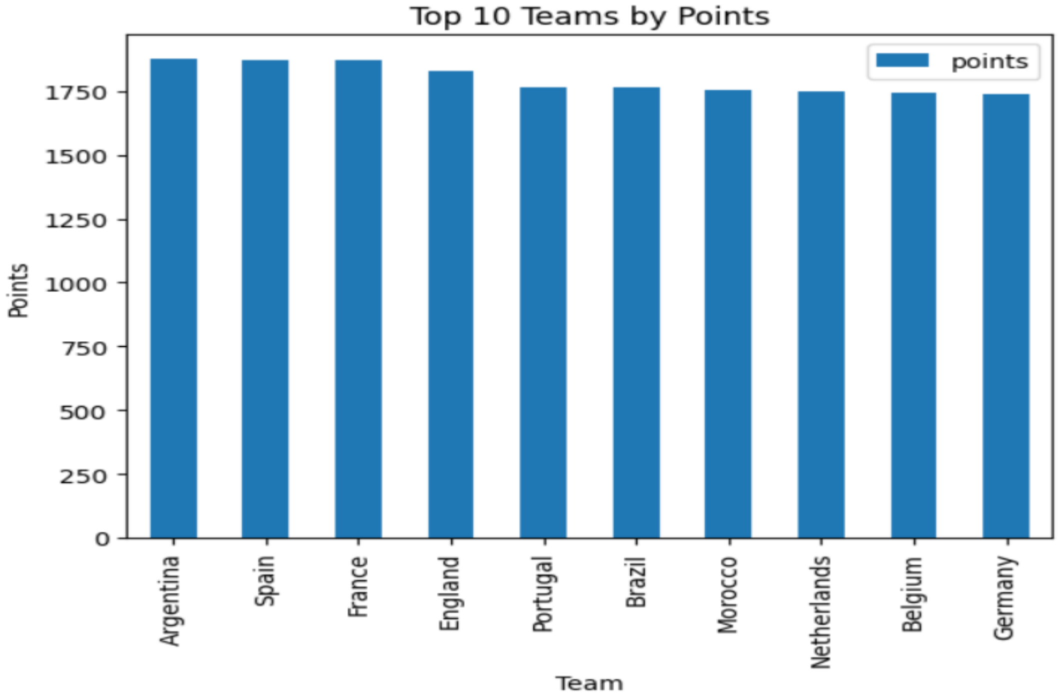
**dtype:** int64

```python
data.describe()
```

| | rank | previous_rank | points | previous_points | rated_matches |
|---|---|---|---|---|---|
| **count** | 211.00000 | 211.00000 | 211.000000 | 211.000000 | 211.000000 |
| **mean** | 106.00000 | 106.00000 | 1229.980217 | 1229.968368 | 45.900474 |
| **std** | 61.05462 | 61.05462 | 278.045705 | 277.802985 | 16.301754 |
| **min** | 1.00000 | 1.00000 | 722.933728 | 726.330601 | 7.000000 |
| **25%** | 53.50000 | 53.50000 | 1010.838116 | 1008.017114 | 35.000000 |
| **50%** | 106.00000 | 106.00000 | 1191.655395 | 1191.416122 | 48.000000 |
| **75%** | 158.50000 | 158.50000 | 1455.740658 | 1455.577884 | 56.000000 |
| **max** | 211.00000 | 211.00000 | 1876.118331 | 1877.322731 | 87.000000 |

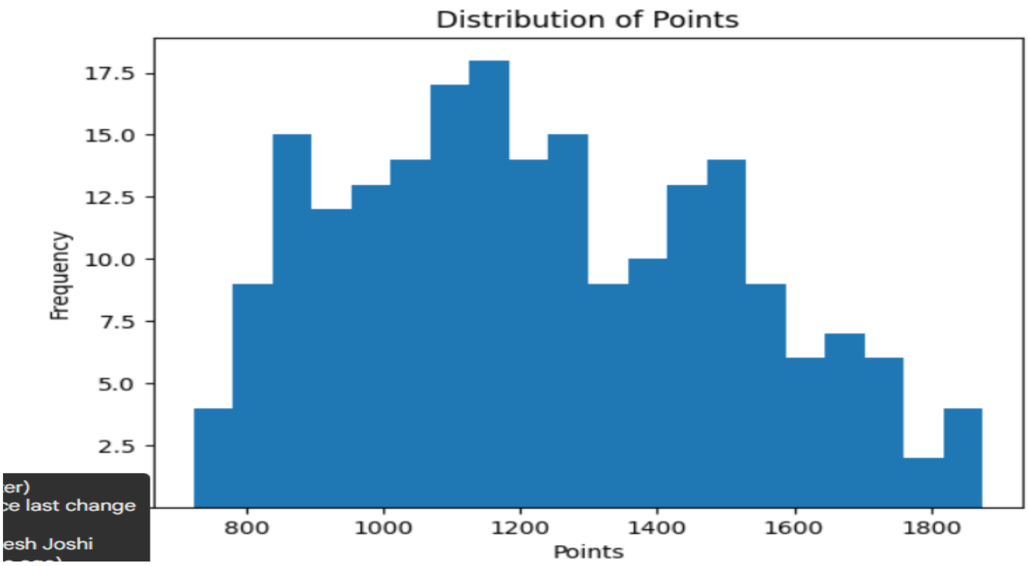
- **bar chart :**

```
data.sort_values('points', ascending=False).head(10).plot(x='team', y='points', kind='bar')
plt.title("Top 10 Teams by Points")
plt.xlabel("Team")
plt.ylabel("Points")
plt.show()
```



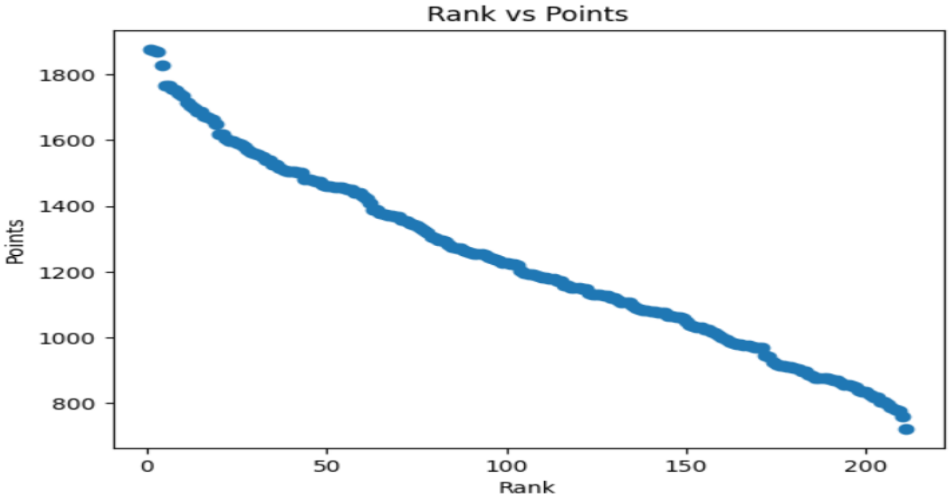
- **Histogram :**

```python
plt.hist(data['points'], bins=20)
plt.title("Distribution of Points")
plt.xlabel("Points")
plt.ylabel("Frequency")
plt.show()
```



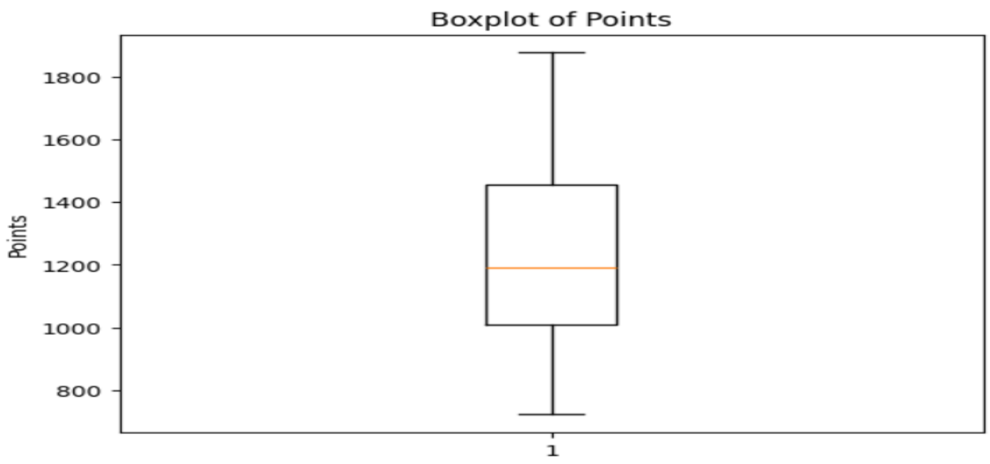
- **scatter plot :**

```
plt.scatter(data['rank'], data['points'])
plt.title("Rank vs Points")
plt.xlabel("Rank")
plt.ylabel("Points")
plt.show()
```



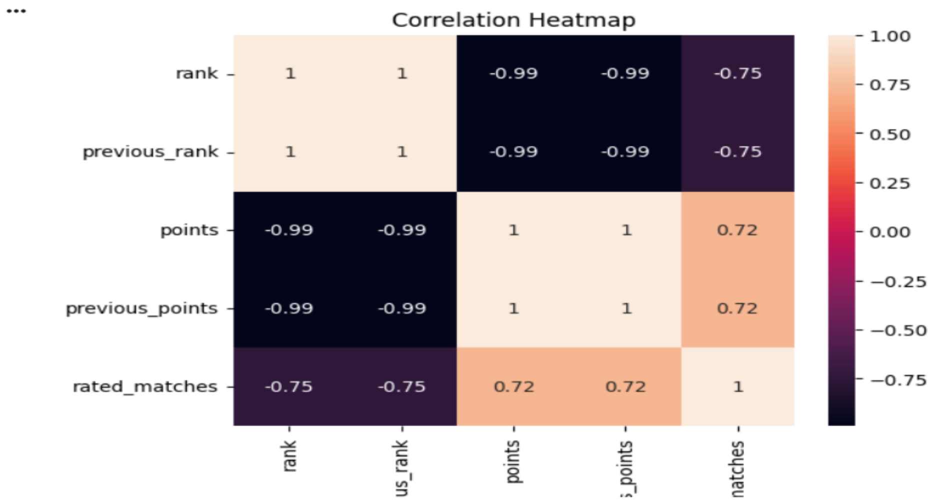
- **box plot :**

```
plt.boxplot(data['points'])
plt.title("Boxplot of Points")
plt.ylabel("Points")
plt.show()
```



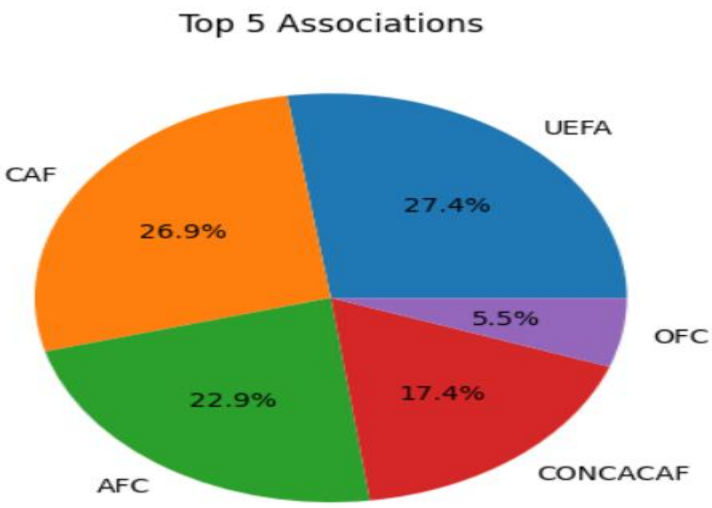
- **Heatmap :**

```
sns.heatmap(data[['rank','previous_rank','points','previous_points','rated_matches']].corr(), annot=True)
plt.title("Correlation Heatmap")
plt.show()
```



- **pie chart :**

```
data['association'].value_counts().head(5).plot.pie(autopct='%1.1f%%')
plt.title("Top 5 Associations")
plt.ylabel("")
plt.show()
```




❖ **Dataset Summary :**

➢ **Columns:** team, team_code, association, rank, previous_rank, points, previous_points, rated_matches
➢ **Type:** Tabular dataset with categorical (team, association) and numerical (rank, points, rated_matches) features.
➢ **Purpose:** Analyze team performance and rankings based on points and matches.

- **Dataset Name:** FIFA World Ranking Dataset.
- Kaggle Dataset Link: - FIFA Football World Cup
- **Dataset Description:** This dataset contains information about international football teams, their rankings, points, associations, and matches. It helps analyze team performance and ranking.
- **Number of Rows and Columns:**
  - Rows:211
  - Columns: 8
- **List of Dataset Features (Columns):**
  - team
  - team_code
  - association
  - rank
  - previous_rank
  - points
  - previous_points
  - rated_matches
- **Input Features (Independent Variables):**
  - team_code
  - association
  - previous_rank
  - previous_points
  - rated_matches
- **Output/Target Variable (Dependent Variable):**
  - rank
- **Conclusion:** Higher points → better ranks, associations differ in representation, and correlation confirms rank and points are closely related.